

FALCUTÉ DES SCIENCES ET TECHNOLOGIES (FST)

TROISIÈME ANNÉE

Rapport du travail de Laboratoire NO 3

Cours : Réseaux1

Étudiante : Roobentia SENECAL

Professeur : Ismaël SAINT AMOUR

LE 05 Novembre 2026

- **Objectif de ce TD**

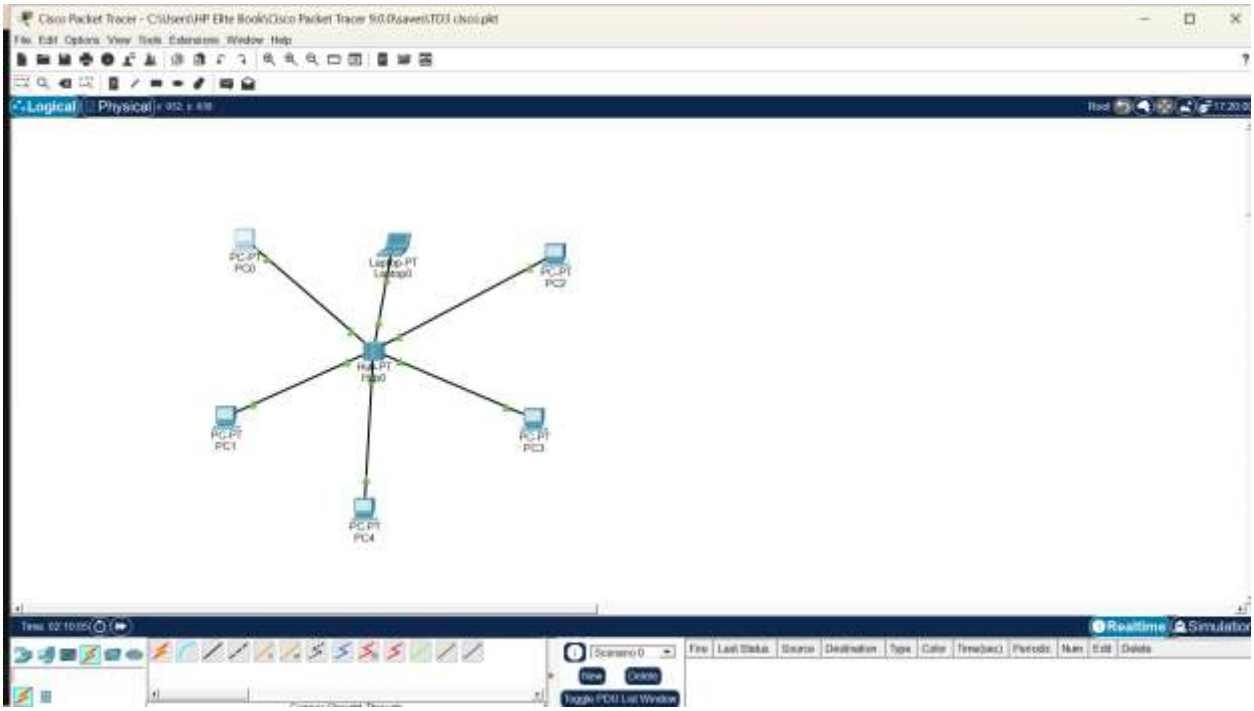
1. Savoir attribuer des adresses IP valides aux machines.

2. Comprendre l'adressage IPv4 et IPv6.

3. Configurer des adresses IP sur des hôtes et routeurs dans Cisco Packet Tracer.

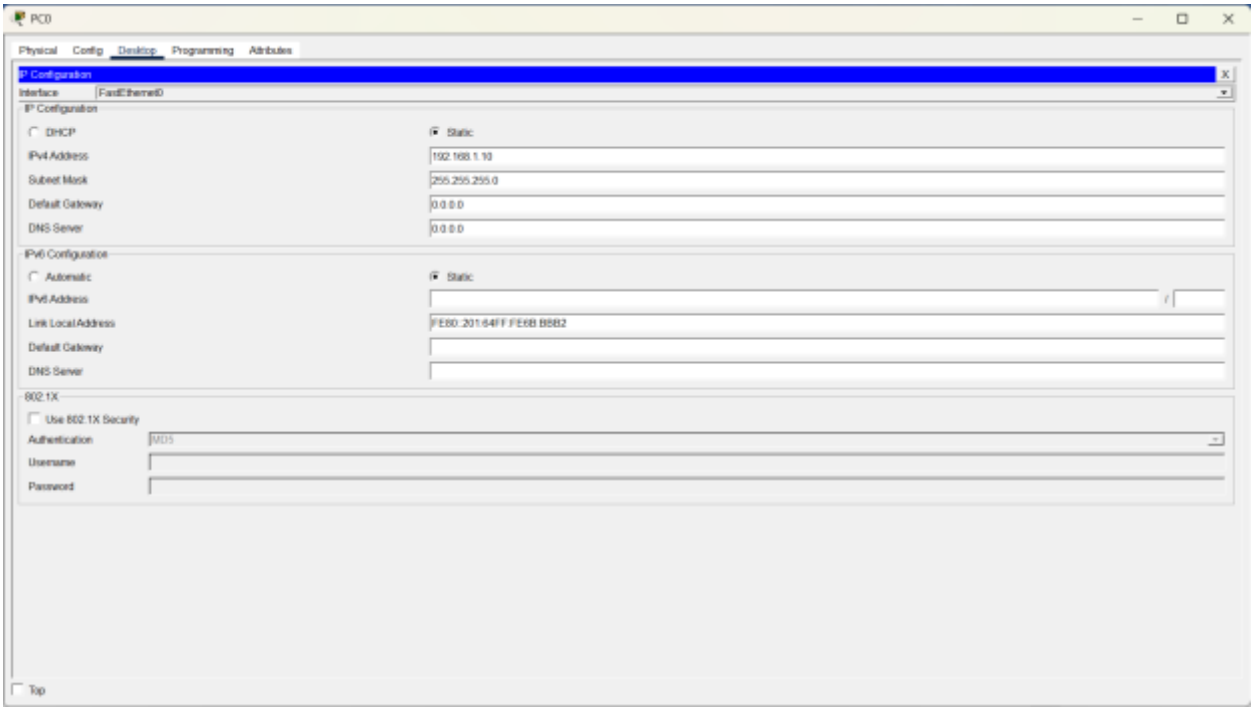
4. Vérifier la connectivité avec les commandes ping et ping ipv6.

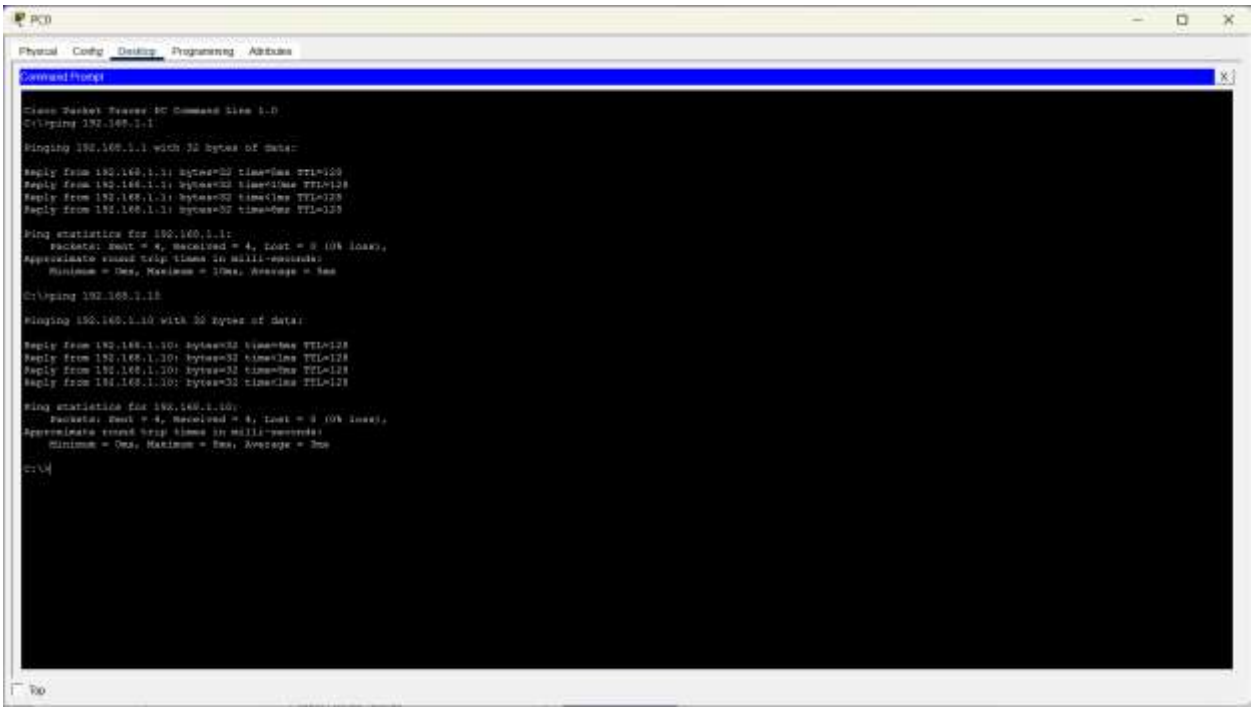
1. Reproduire une topologie que le professeur a donné en configurant les adresses IPv4 indiquées dans le tableau donné, puis vérifiez la connectivité.



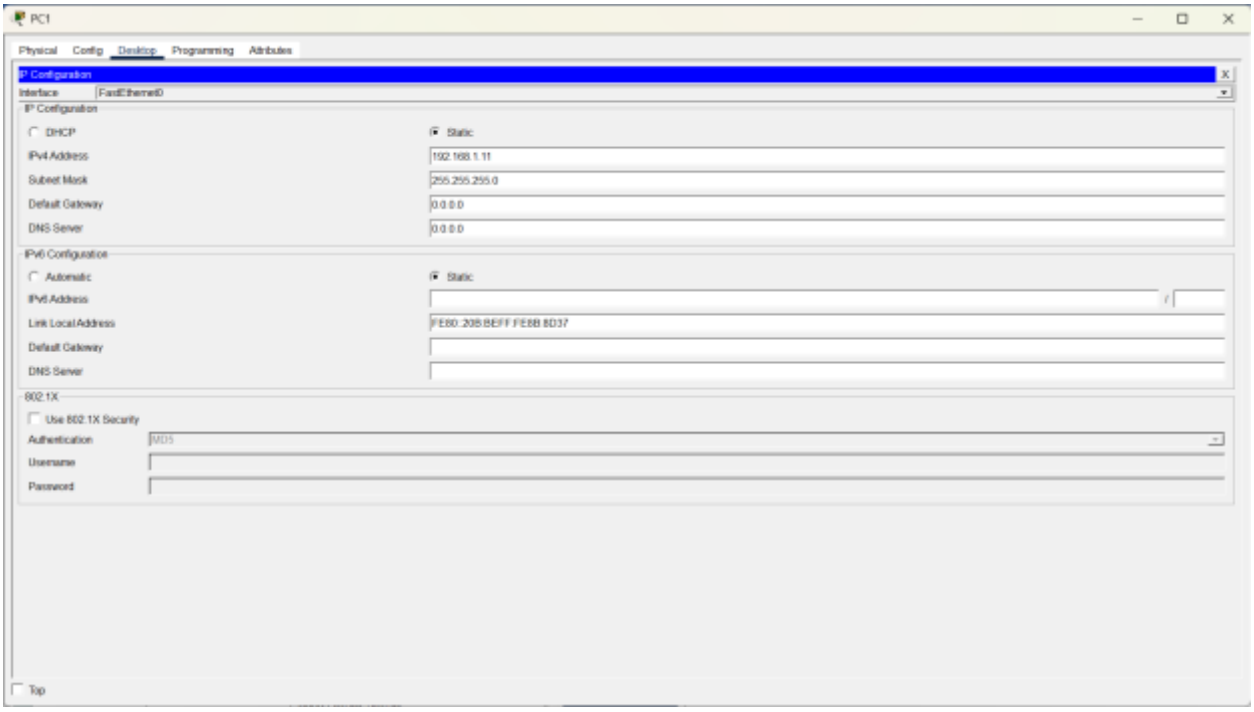
Verification des adresses IP :

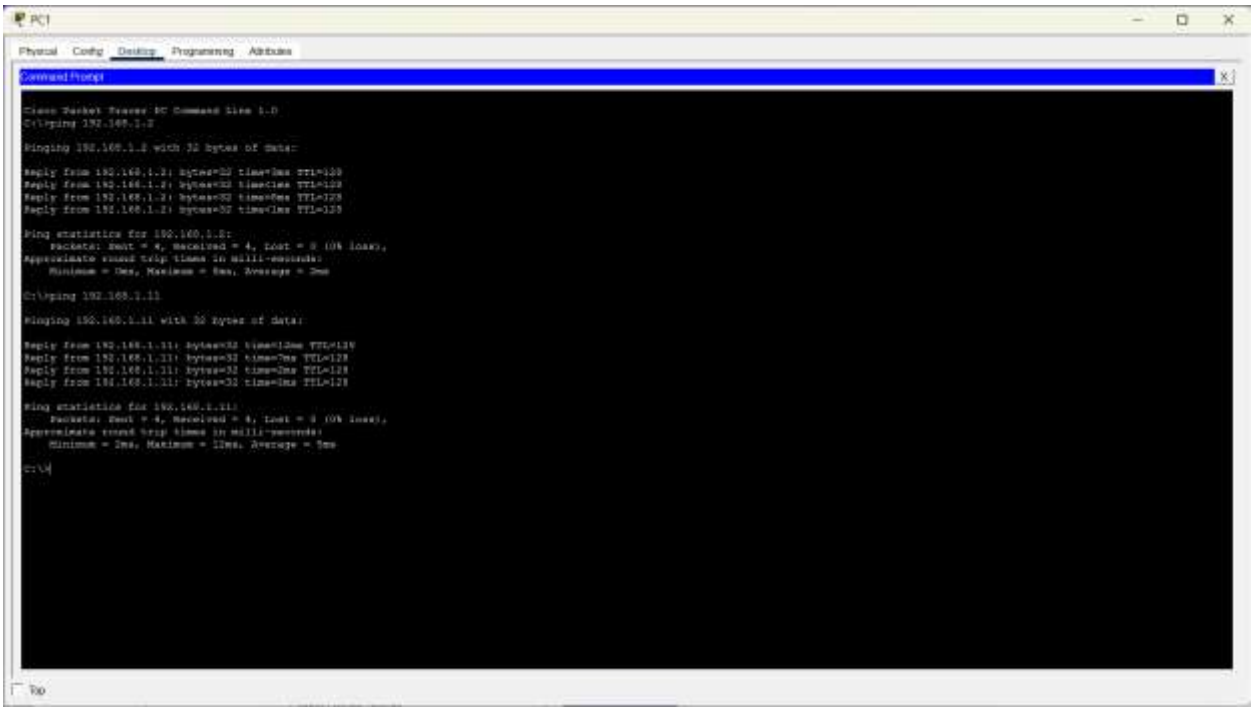
1-



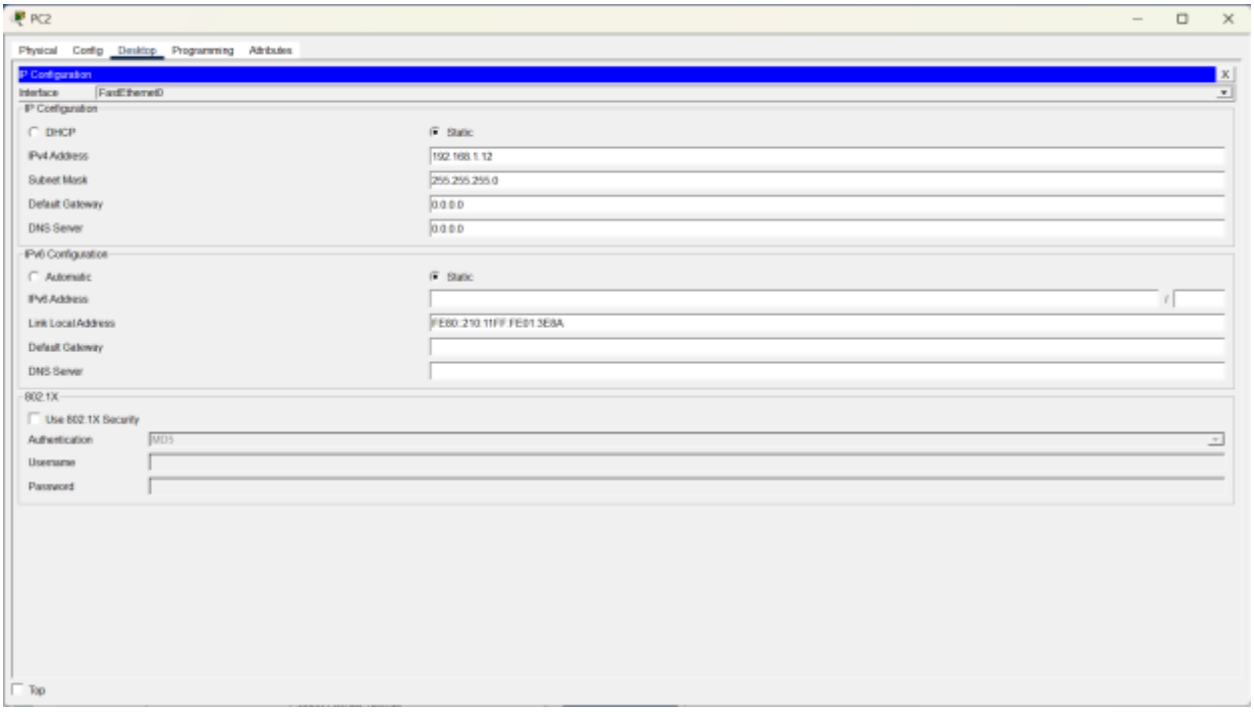

PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=0ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=0ms TTL=128
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=0ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=0ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top

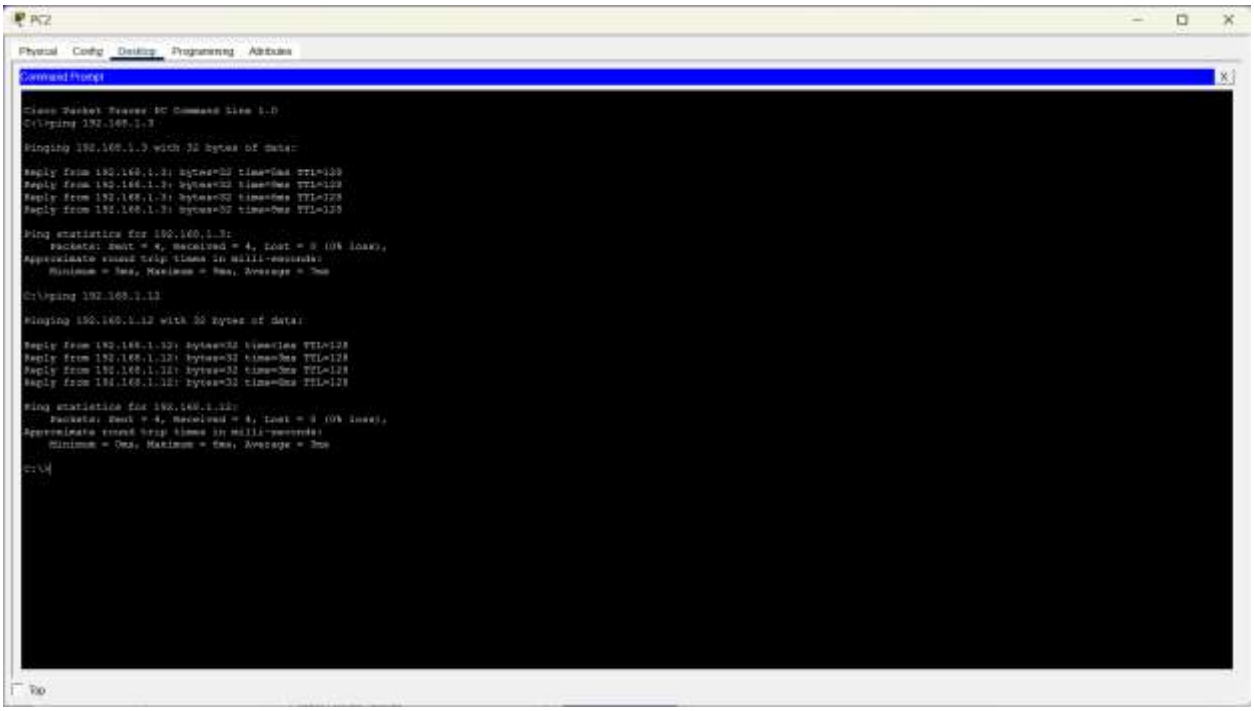
2-


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20B:BEFF:FE8B:8D37
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

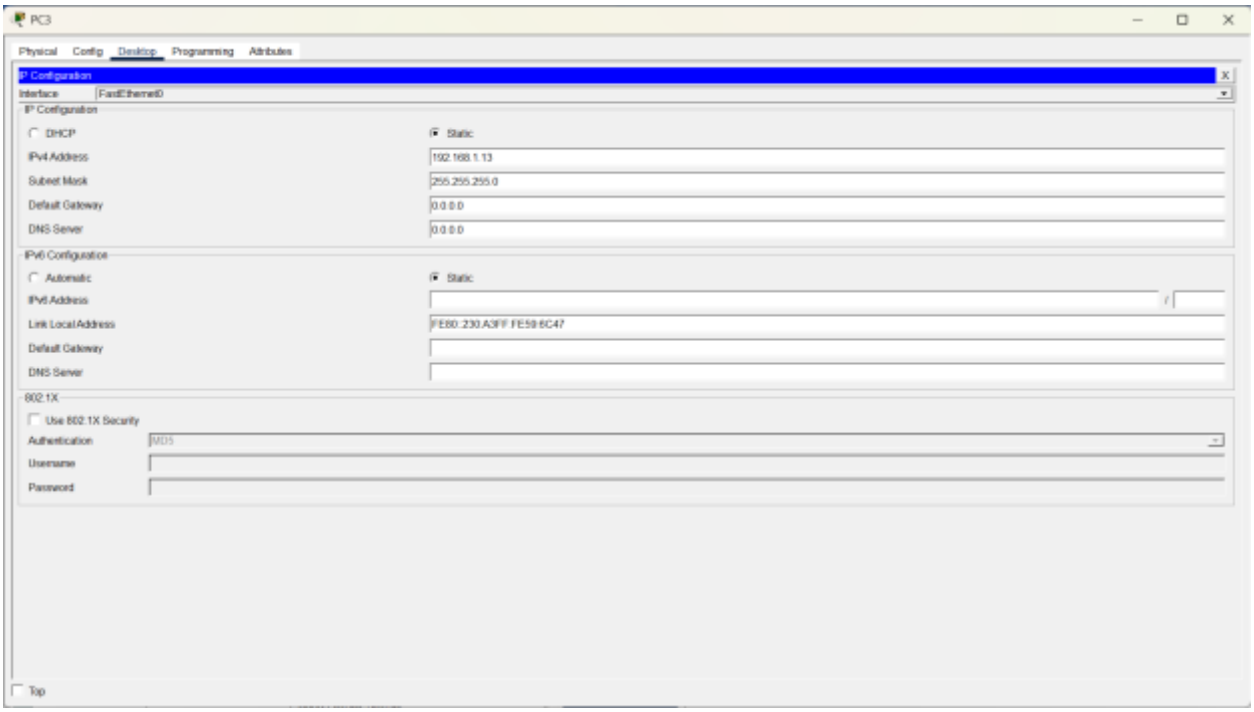

PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=0ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=0ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=0ms TTL=128
Reply from 192.168.1.11: bytes=32 time=0ms TTL=128
Reply from 192.168.1.11: bytes=32 time=0ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

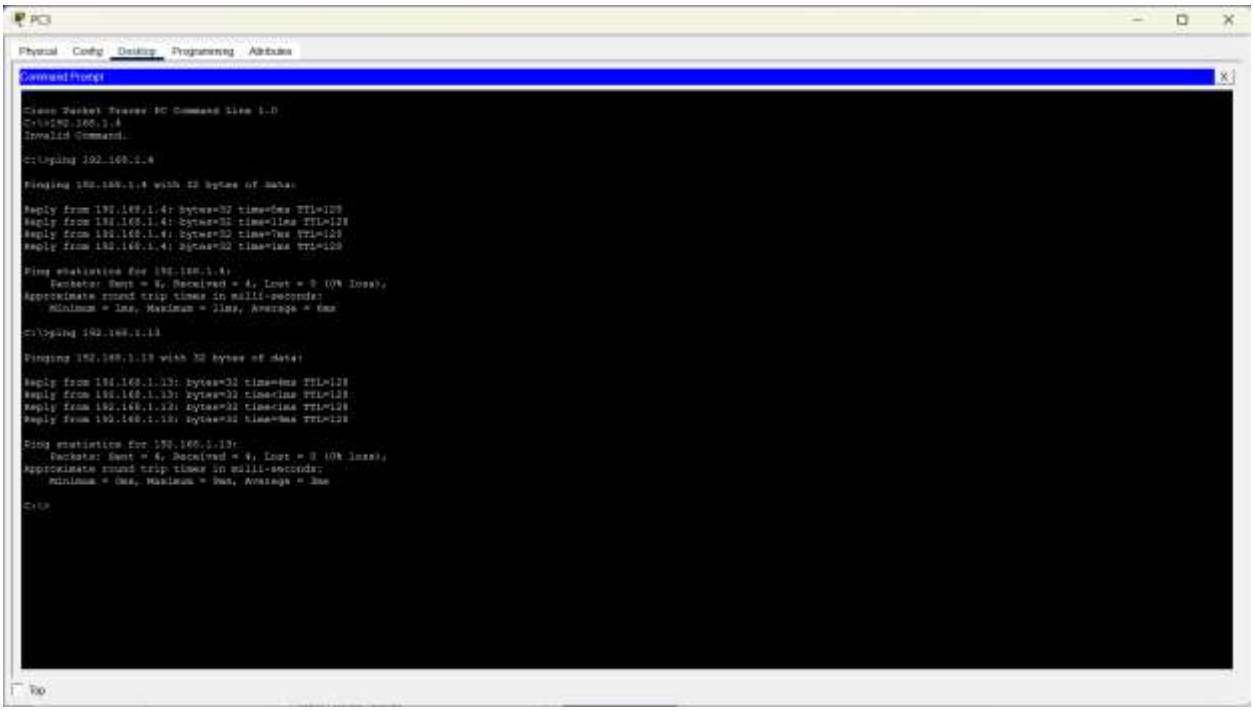
3-


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.12
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::210:11FF:FE61:3E8A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

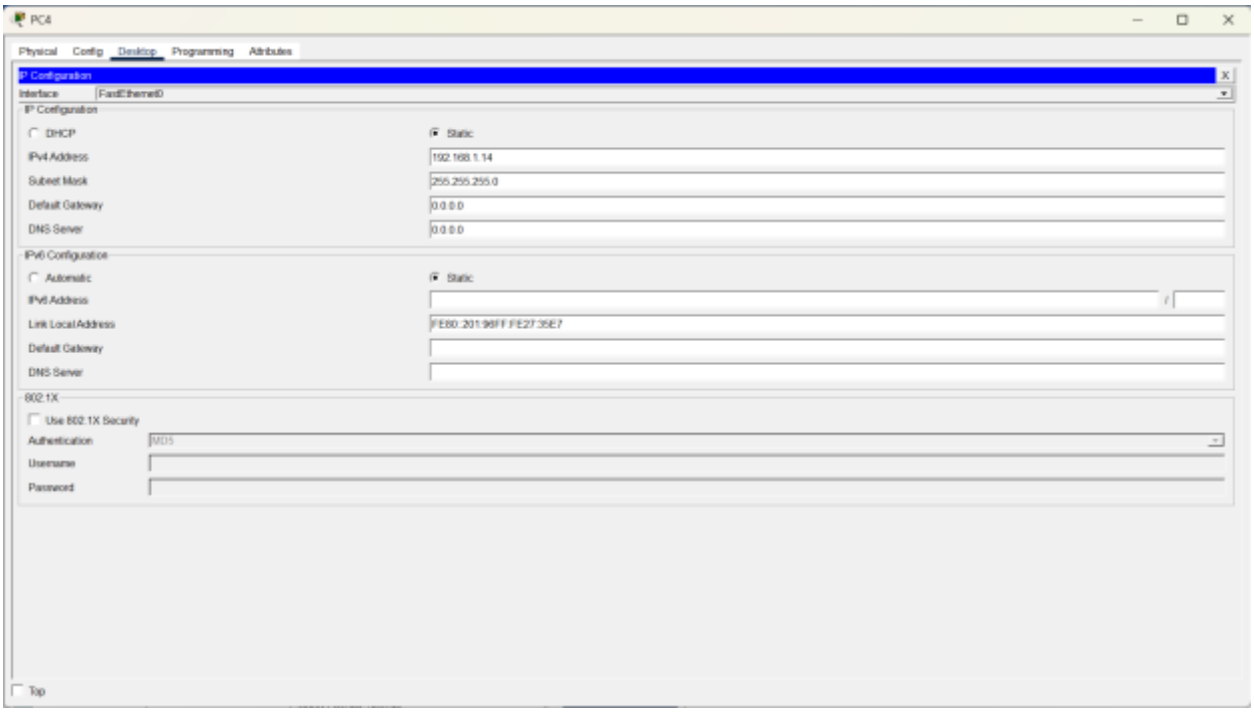

PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=0ms TTL=128
Reply from 192.168.1.3: bytes=32 time=0ms TTL=128
Reply from 192.168.1.3: bytes=32 time=0ms TTL=128
Reply from 192.168.1.3: bytes=32 time=0ms TTL=128
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top

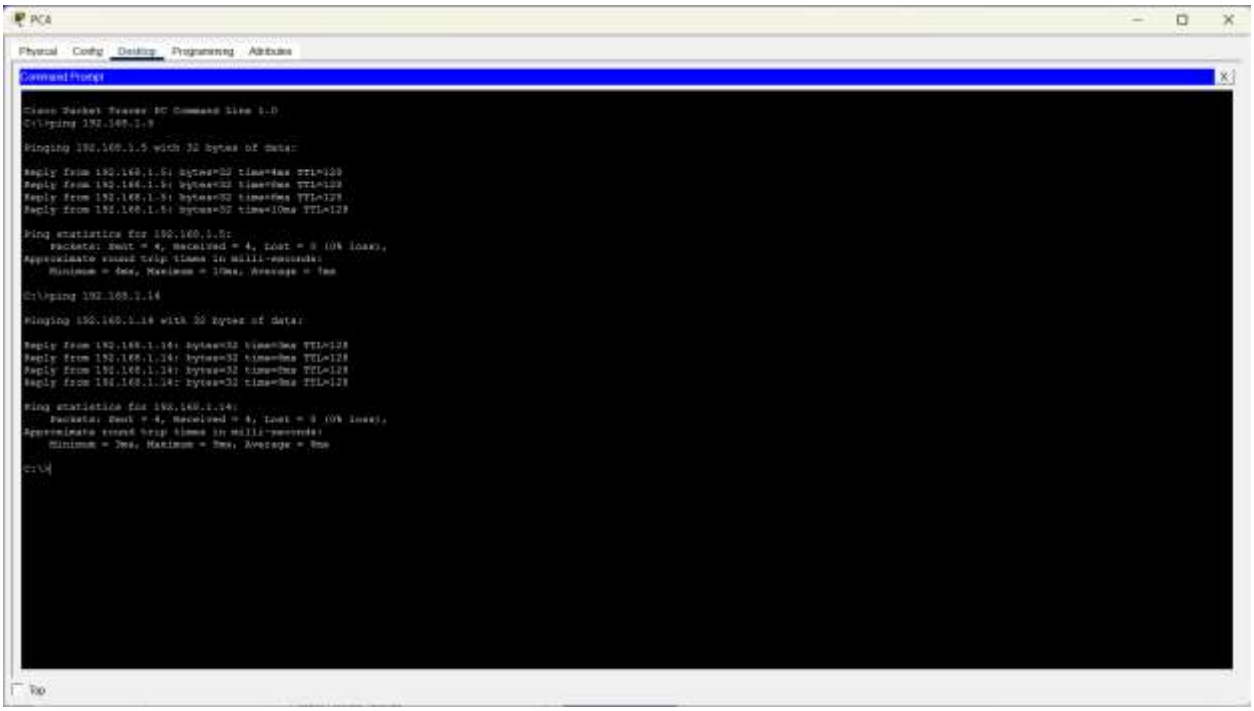
4-


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.13
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::230:A3FF:FE59:6C47
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

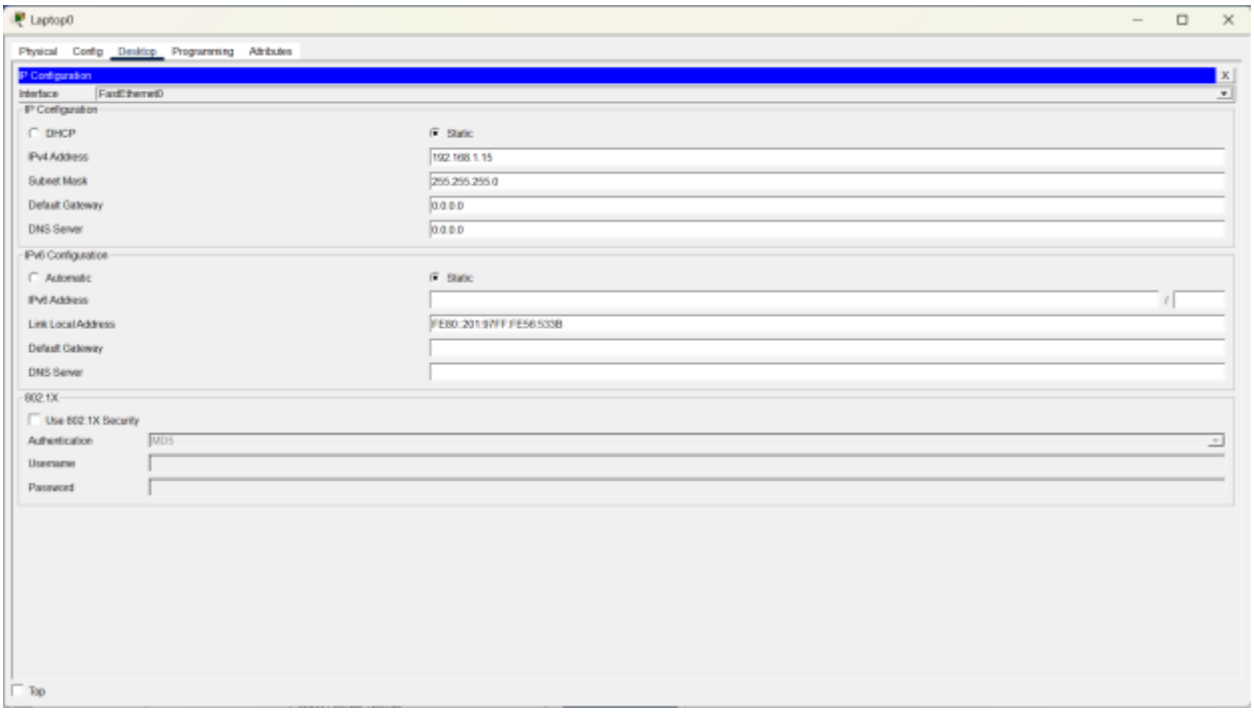

PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>192.168.1.4
Invalid Command.
C:\>ping 192.168.1.4
Pinging 192.168.1.4 with 32 bytes of data:
Reply from 192.168.1.4: bytes=32 time=0ms TTL=128
Reply from 192.168.1.4: bytes=32 time=11ms TTL=128
Reply from 192.168.1.4: bytes=32 time=7ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 11ms, Average = 6ms
C:\>ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time=0ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=0ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top

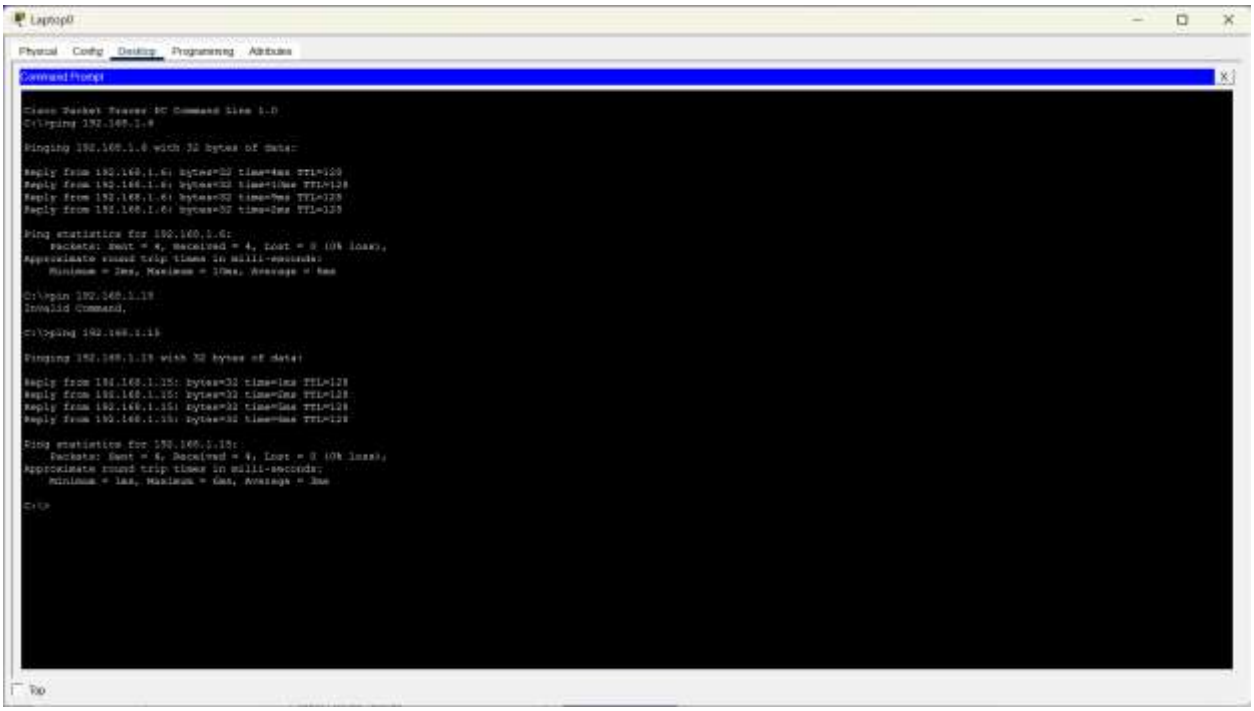
5-


PC4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.14
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:96FF:FE27:35E7
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC4
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.5
Pinging 192.168.1.5 with 32 bytes of data:
Reply from 192.168.1.5: bytes=32 time=4ms TTL=128
Reply from 192.168.1.5: bytes=32 time=0ms TTL=128
Reply from 192.168.1.5: bytes=32 time=0ms TTL=128
Reply from 192.168.1.5: bytes=32 time=10ms TTL=128
Ping statistics for 192.168.1.5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 3ms
C:\>ping 192.168.1.14
Pinging 192.168.1.14 with 32 bytes of data:
Reply from 192.168.1.14: bytes=32 time=0ms TTL=128
Reply from 192.168.1.14: bytes=32 time=0ms TTL=128
Reply from 192.168.1.14: bytes=32 time=0ms TTL=128
Reply from 192.168.1.14: bytes=32 time=0ms TTL=128
Ping statistics for 192.168.1.14:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top

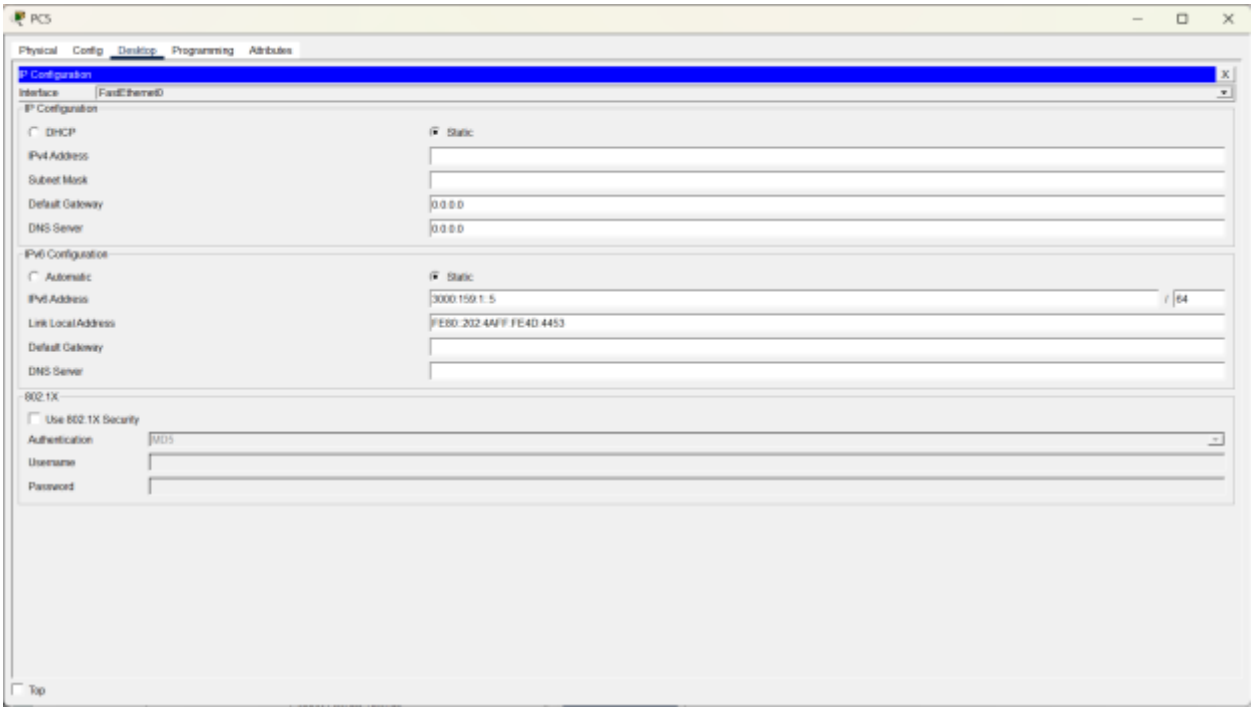
6-

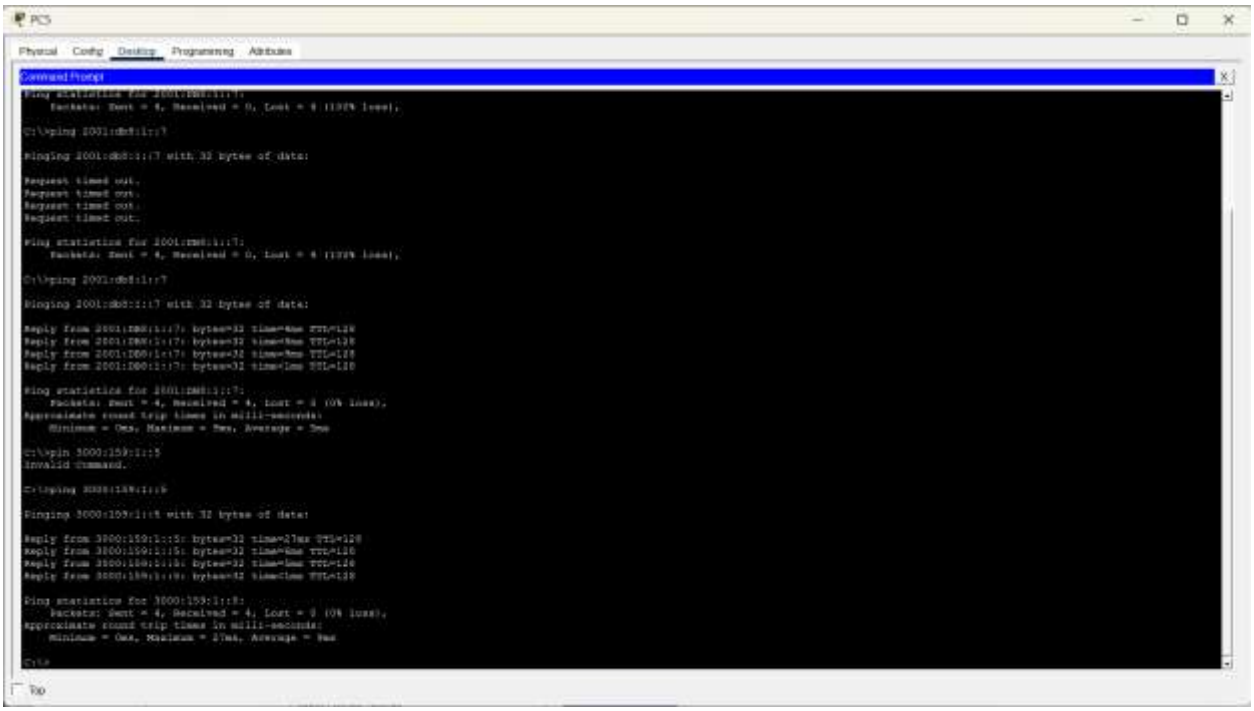

Laptop0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.15
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:97FF:FE56:533B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top



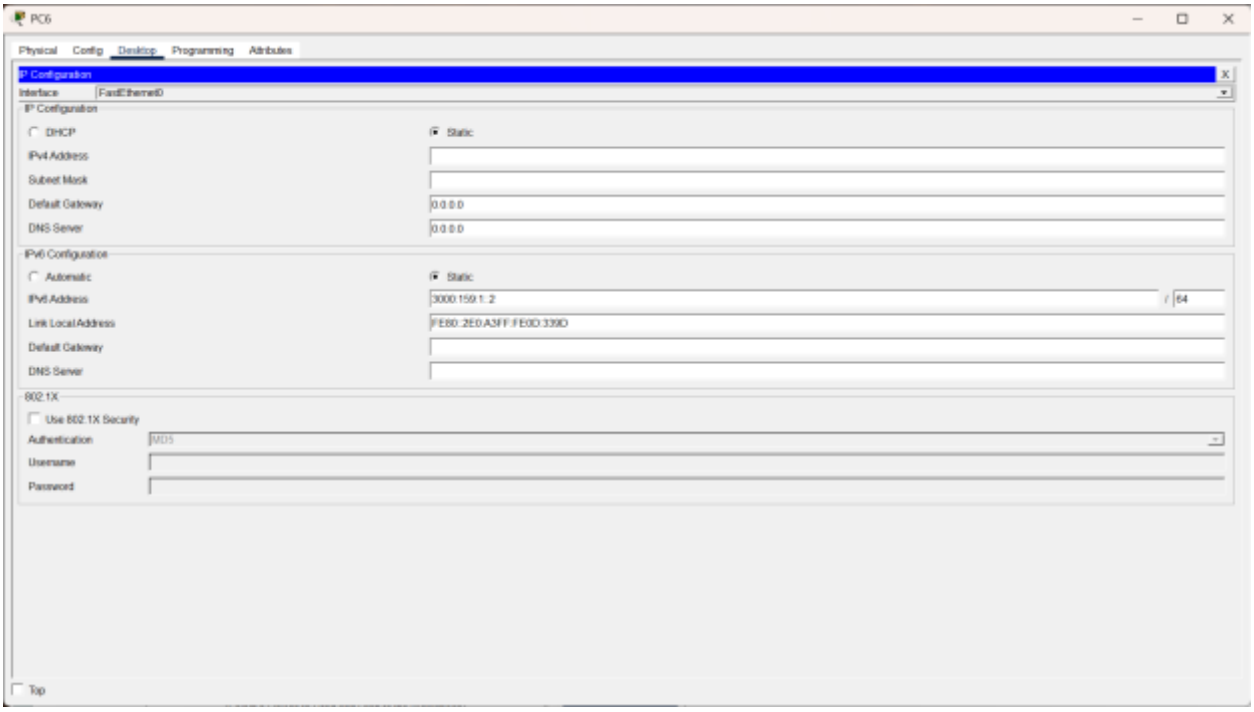
2- 5. Reproduisez une topologie en configurant les adresses IPv6 indiquées dans le tableau donné, puis vérifiez la connectivité.

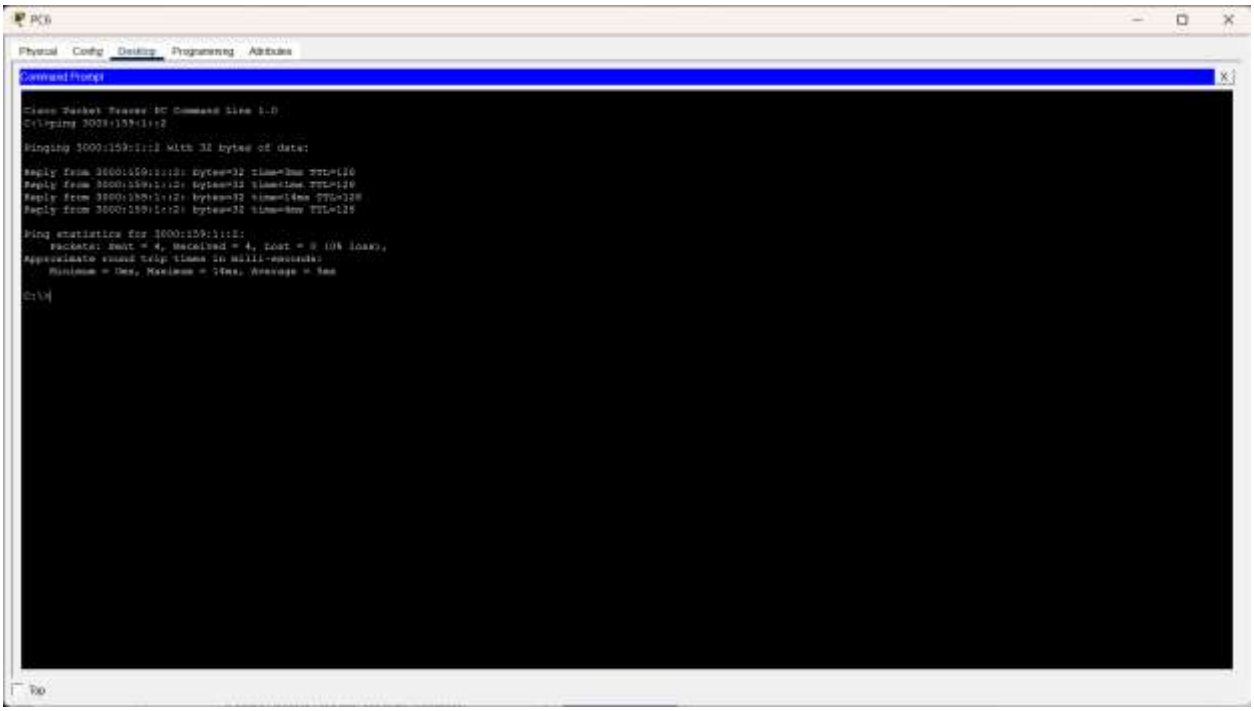
1-



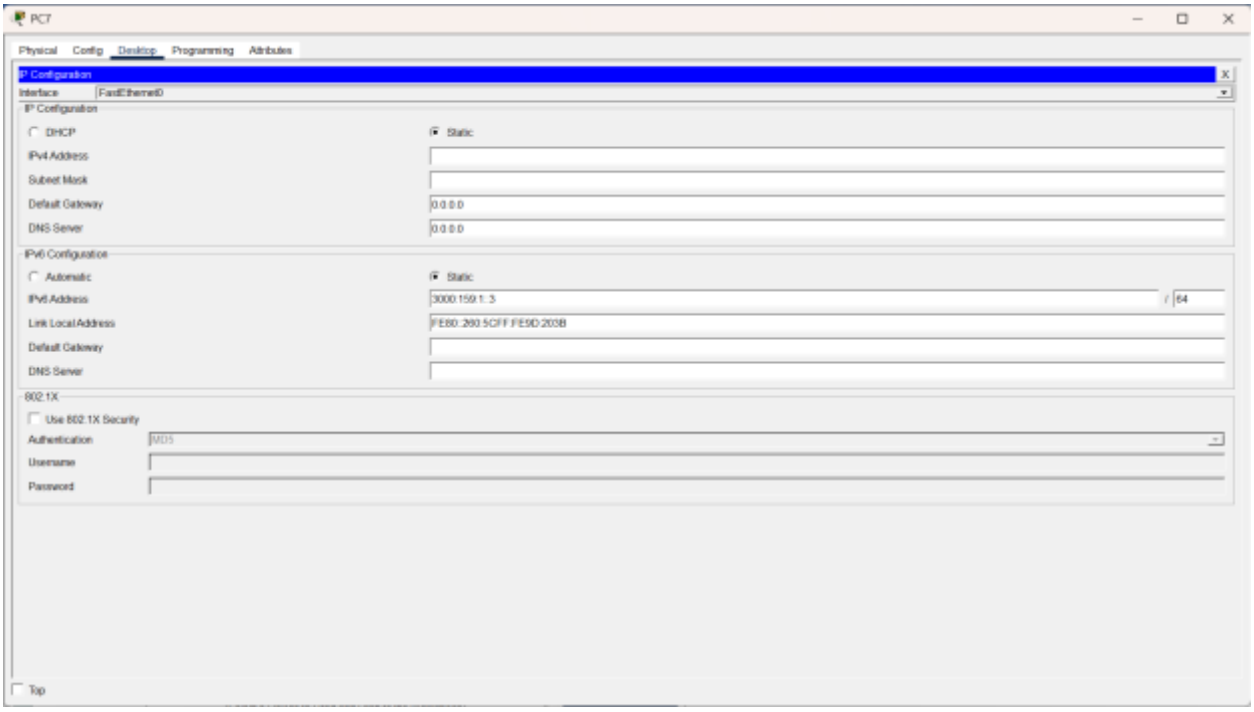

PC5
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Ping statistics for 2001:DB8:1::7:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 2001:db8:1::7
Pinging 2001:db8:1::7 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 2001:DB8:1::7:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 2001:db8:1::7
Pinging 2001:db8:1::7 with 32 bytes of data:
Reply from 2001:DB8:1::7: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::7: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::7: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::7: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::7:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 3ms
C:\>ping 3000:159:1::5
Invalid Command.
C:\>ping 3000:159:1::5
Pinging 3000:159:1::5 with 32 bytes of data:
Reply from 3000:159:1::5: bytes=32 time=27ms TTL=128
Reply from 3000:159:1::5: bytes=32 time=6ms TTL=128
Reply from 3000:159:1::5: bytes=32 time=1ms TTL=128
Reply from 3000:159:1::5: bytes=32 time=1ms TTL=128
Ping statistics for 3000:159:1::5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 27ms, Average = 9ms
C:\>
Top

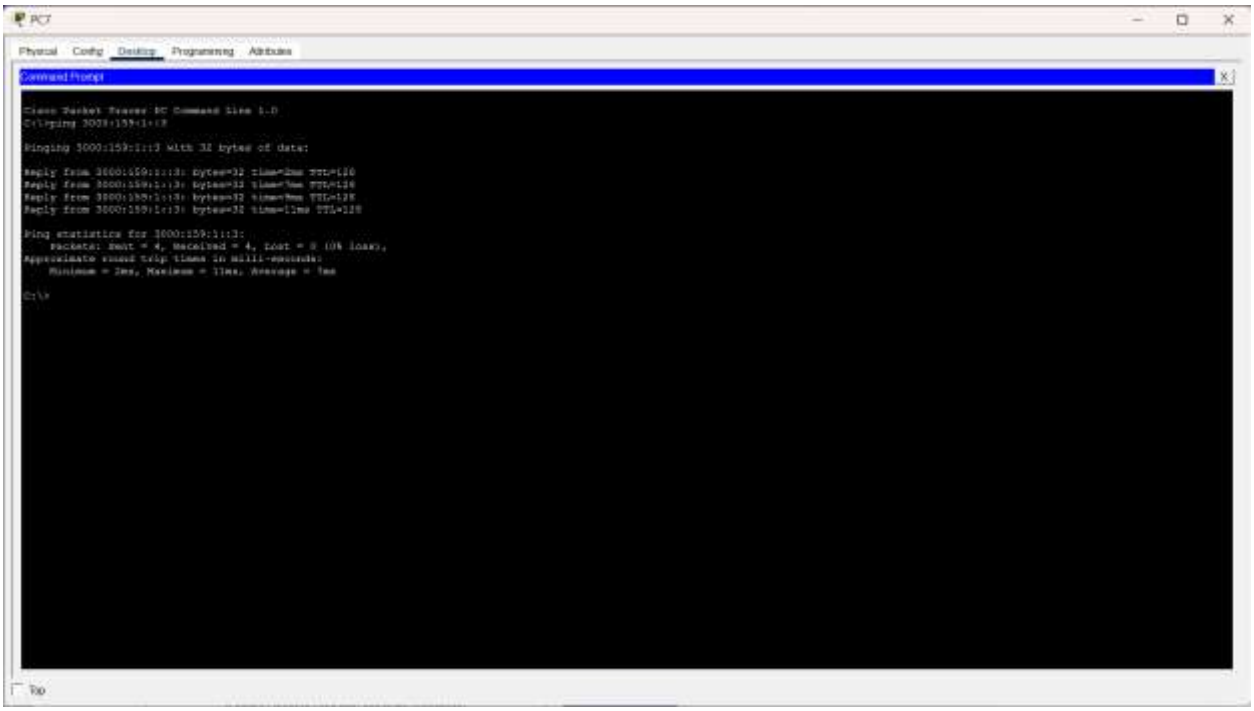
2-


PC6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::2
/ 64
Link Local Address
FE80::2E0:A3FF:FE0D:339D
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

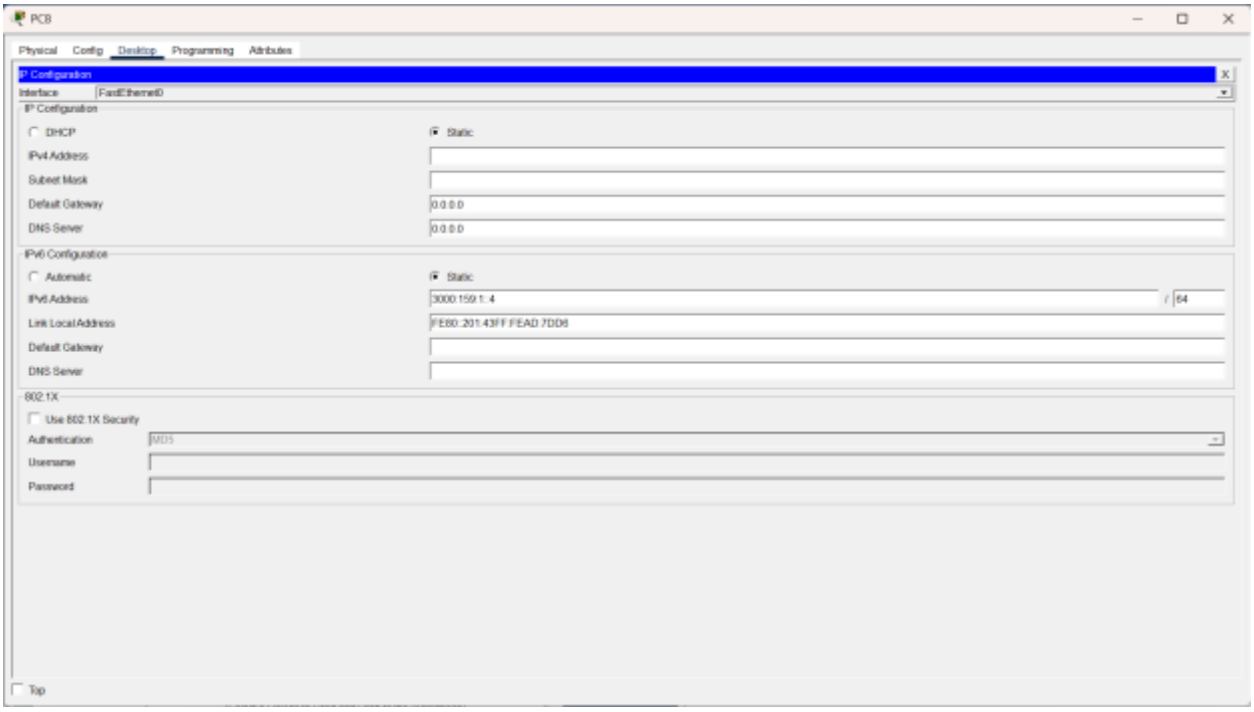

PC6
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::2
Pinging 3000:159:1::2 with 32 bytes of data:
Reply from 3000:159:1::2: bytes=32 time=0ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=14ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=0ms TTL=128
Ping statistics for 3000:159:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 14ms, Average = 3ms
C:\>
Top

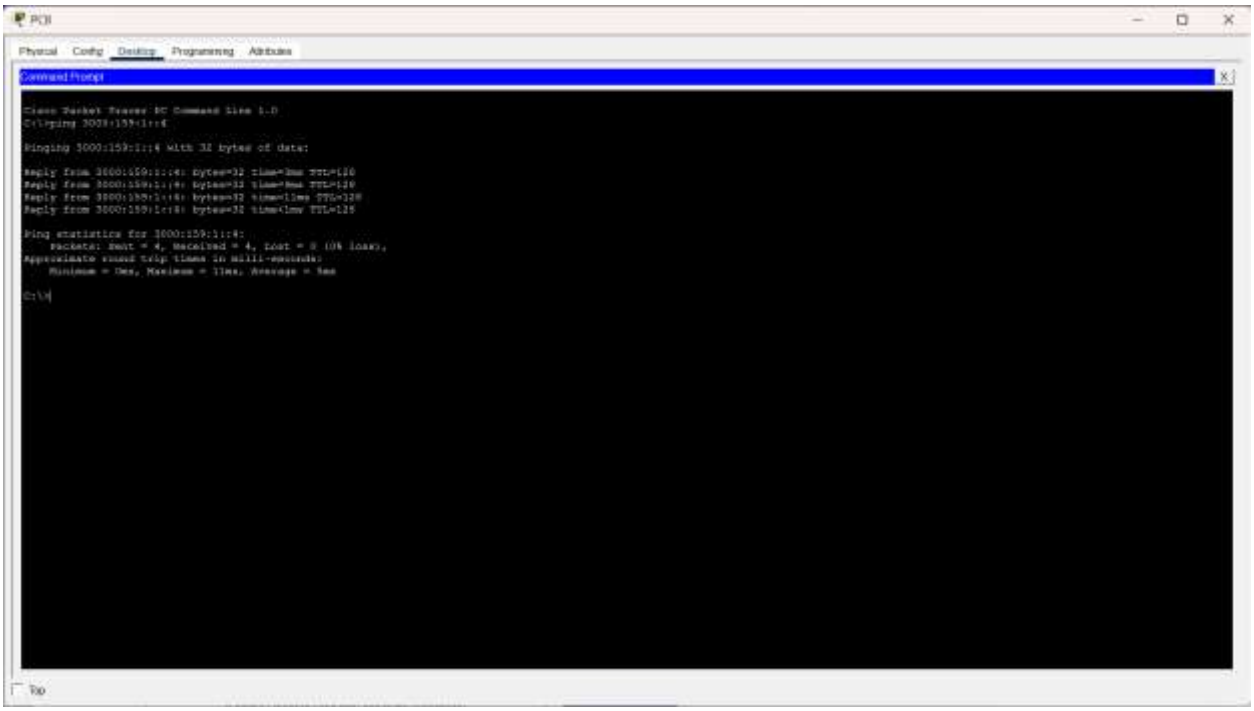
3-


PC7
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::3
/ 64
Link Local Address
FE80::260:5CFF:FE5D:203B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

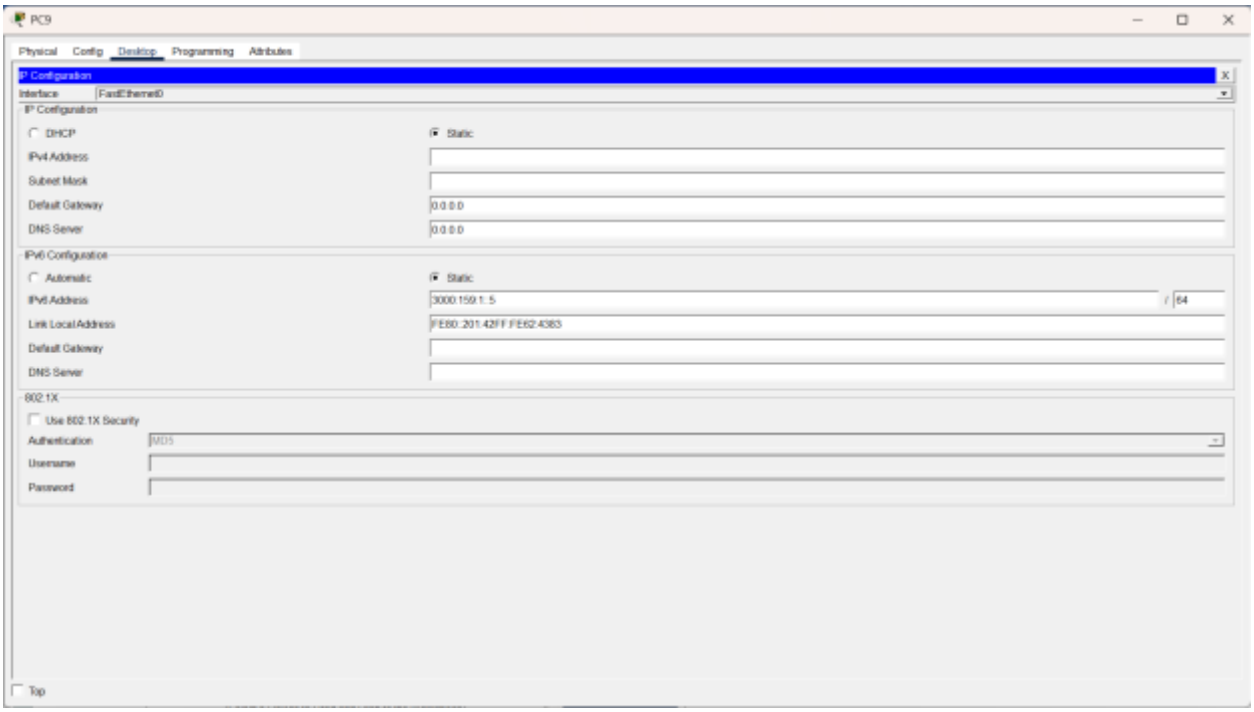

PC7
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::3
Pinging 3000:159:1::3 with 32 bytes of data:
Reply from 3000:159:1::3: bytes=32 time=2ms TTL=126
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=126
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=126
Reply from 3000:159:1::3: bytes=32 time=11ms TTL=126
Ping statistics for 3000:159:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 11ms, Average = 5ms
C:\>
Top

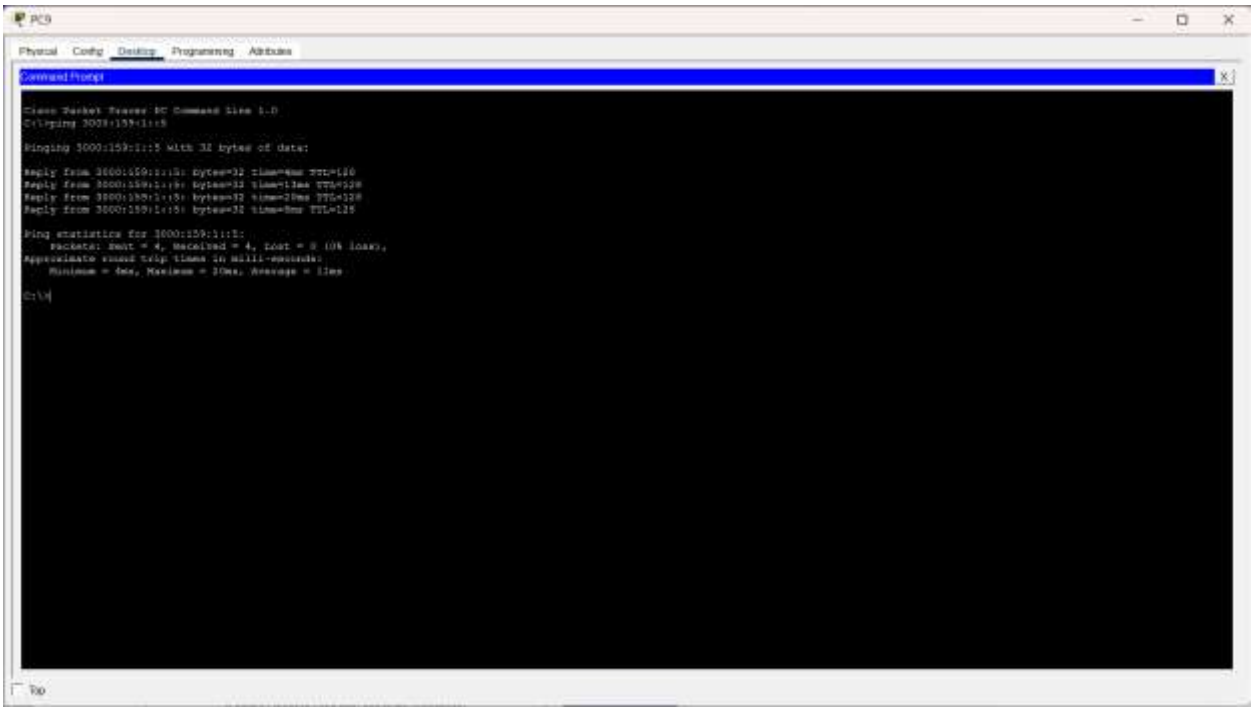
4-


PC8
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::4
/ 64
Link Local Address
FE80::201:43FF:FEAD:7DD6
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

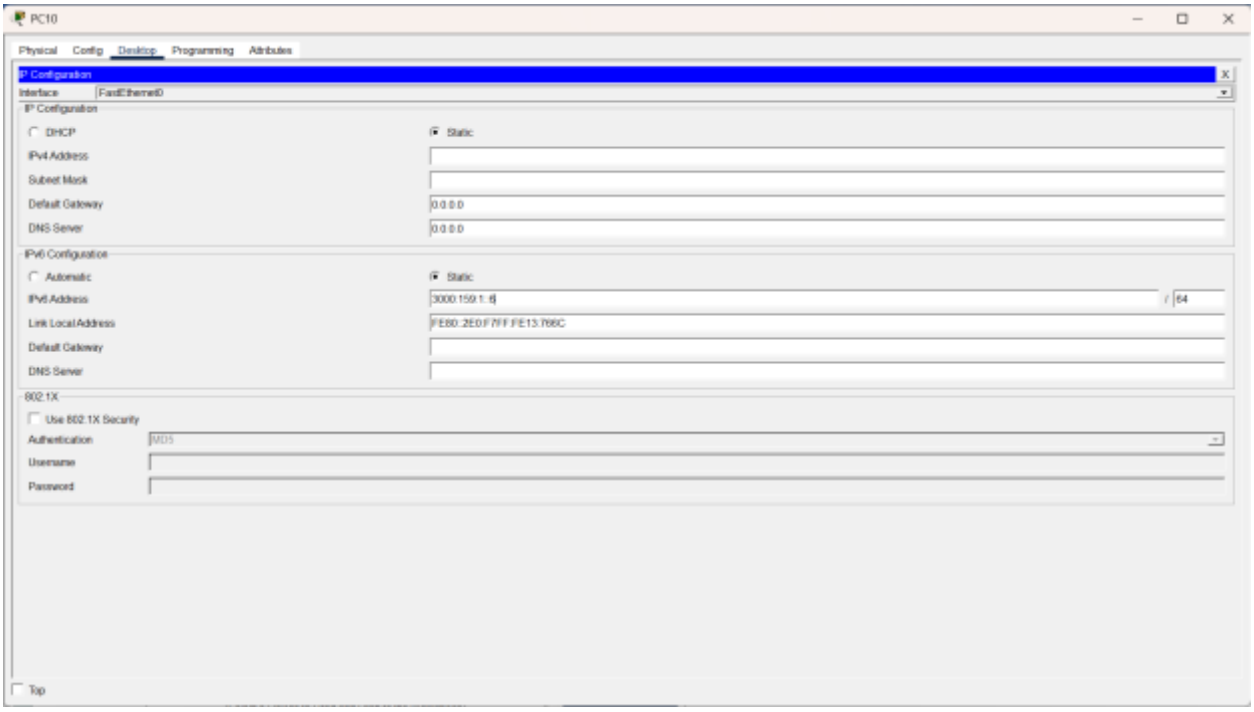

PC8
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::4
Pinging 3000:159:1::4 with 32 bytes of data:
Reply from 3000:159:1::4: bytes=32 time=0ms TTL=128
Reply from 3000:159:1::4: bytes=32 time=0ms TTL=128
Reply from 3000:159:1::4: bytes=32 time=11ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 3000:159:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 3ms
C:\>
Top

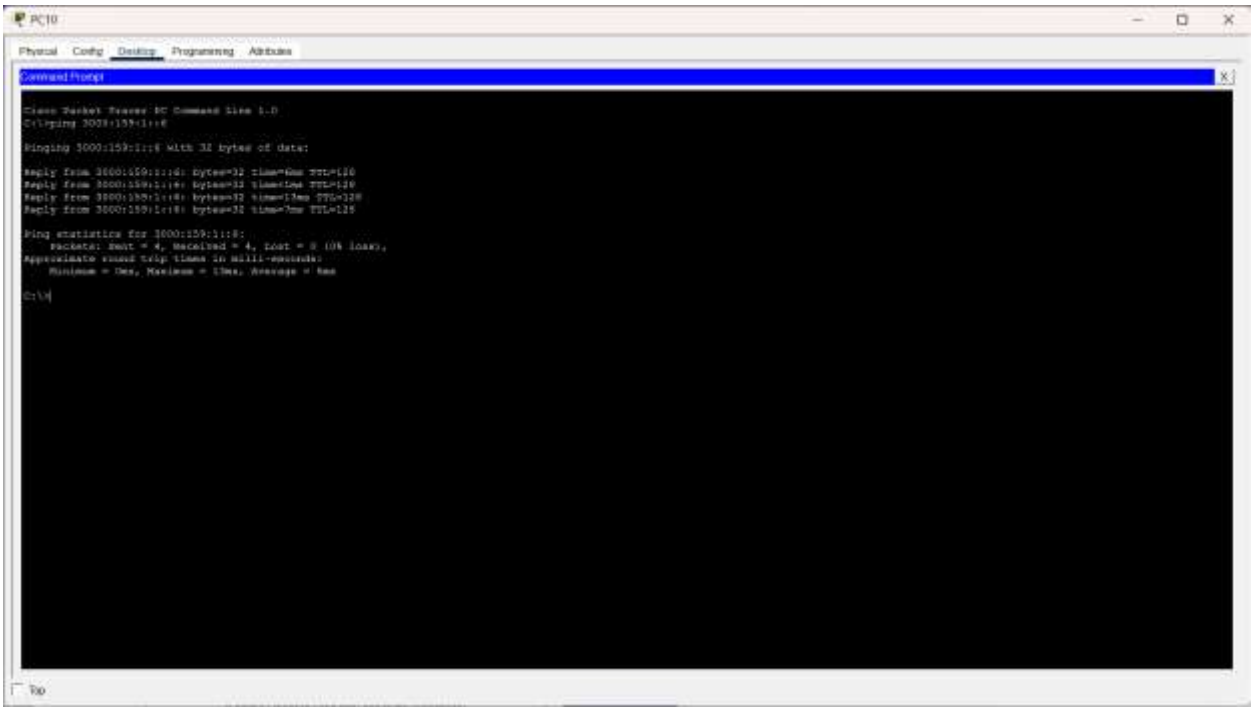
5-


PC9
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::5
/ 64
Link Local Address
FE80::201:42FF:FE62:4383
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

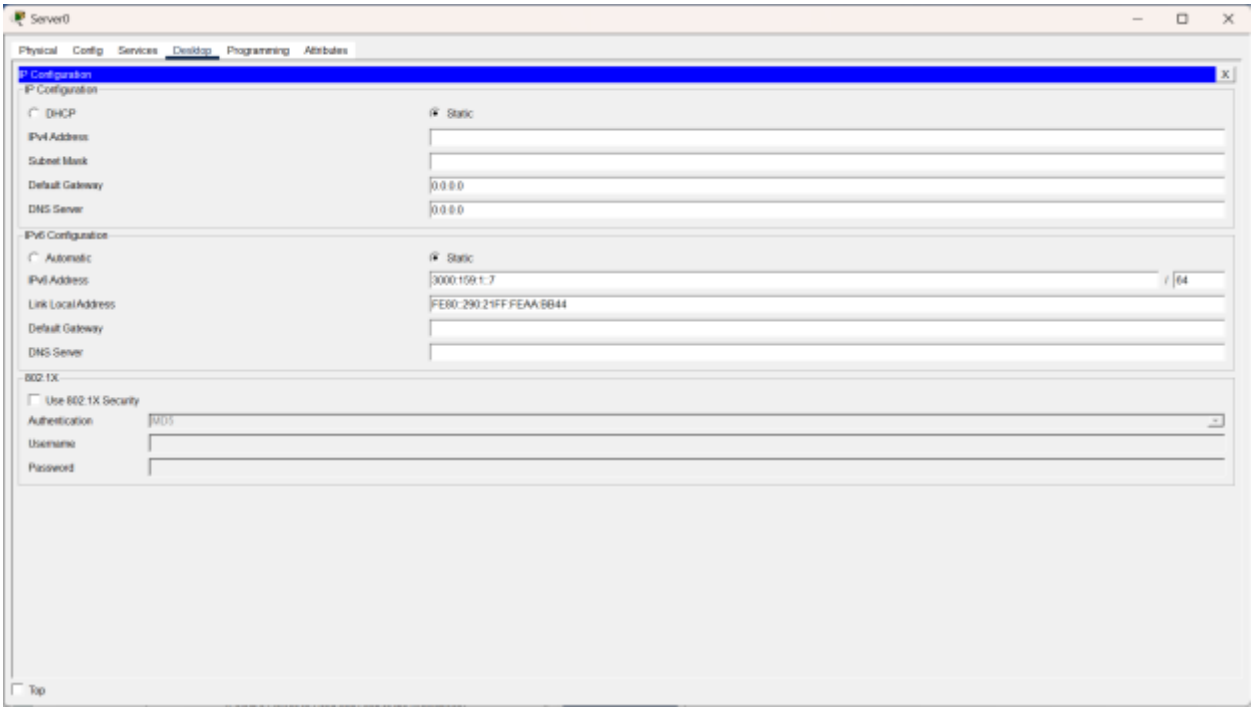

PC9
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::5
Pinging 3000:159:1::5 with 32 bytes of data:
Reply from 3000:159:1::5: bytes=32 time=4ms TTL=128
Reply from 3000:159:1::5: bytes=32 time=11ms TTL=128
Reply from 3000:159:1::5: bytes=32 time=20ms TTL=128
Reply from 3000:159:1::5: bytes=32 time=0ms TTL=128
Ping statistics for 3000:159:1::5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 20ms, Average = 11ms
C:\>
Top

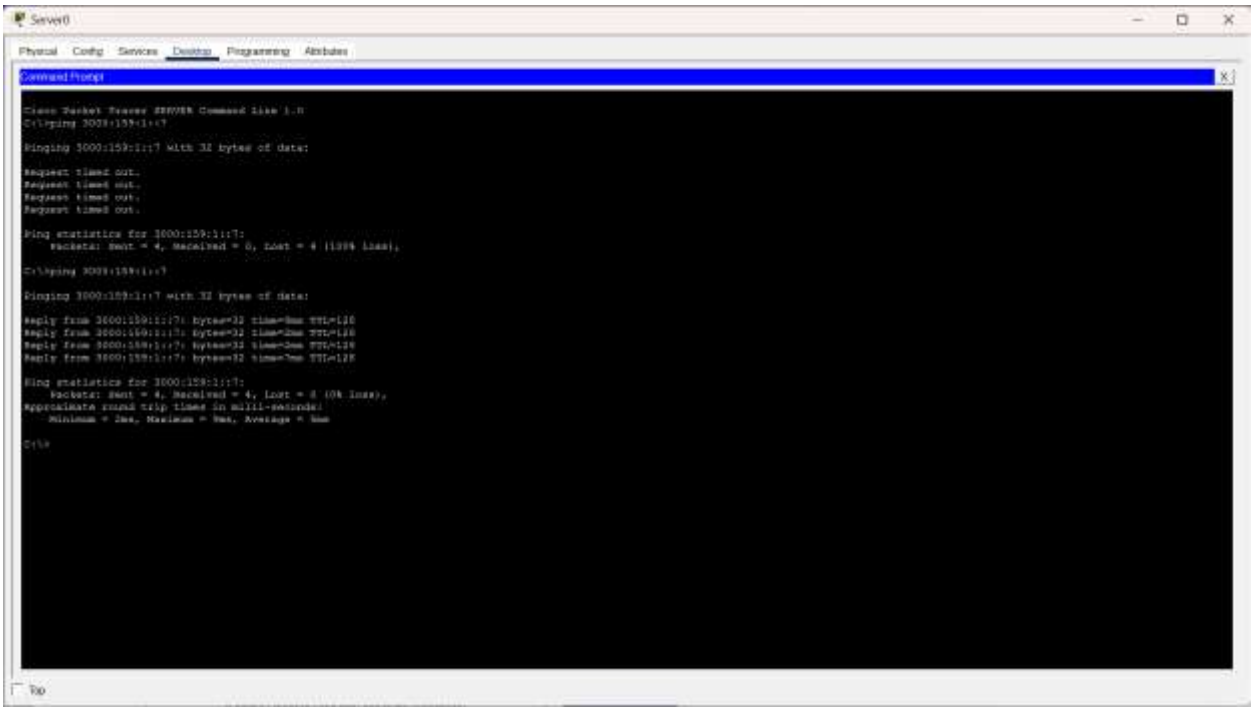
6-


PC10
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::6
/ 64
Link Local Address
FE80::2E0:F7FF:FE13:766C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

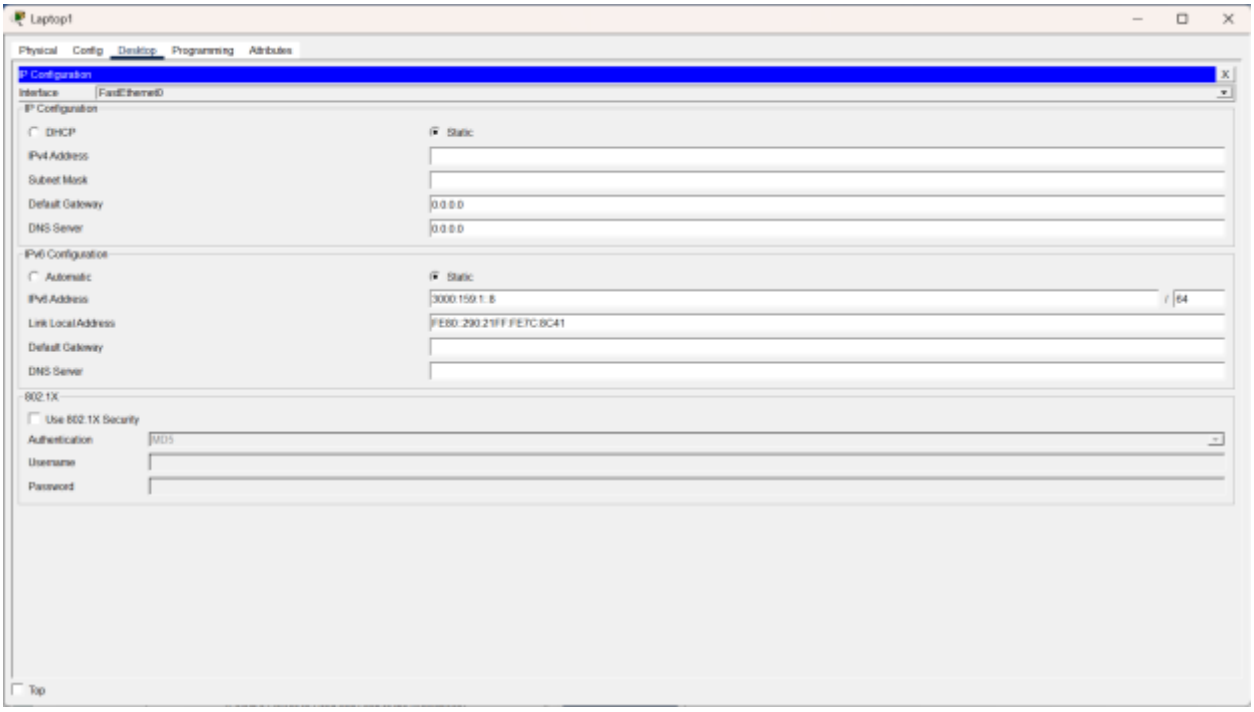

PC10
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::6
Pinging 3000:159:1::6 with 32 bytes of data:
Reply from 3000:159:1::6: bytes=32 time=6ms TTL=125
Reply from 3000:159:1::6: bytes=32 time<1ms TTL=125
Reply from 3000:159:1::6: bytes=32 time=17ms TTL=125
Reply from 3000:159:1::6: bytes=32 time=7ms TTL=125
Ping statistics for 3000:159:1::6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 17ms, Average = 8ms
C:\>
Top

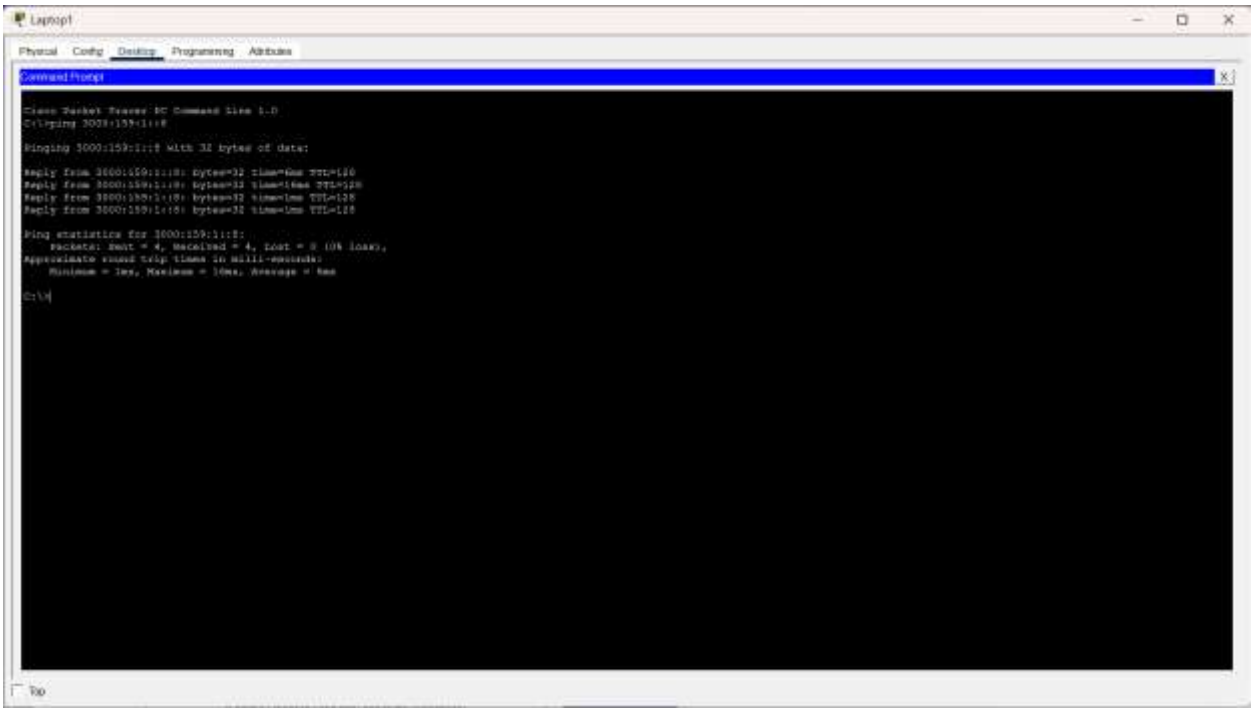
7-


Server0
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::7
/ 64
Link Local Address
FE80::290:21FF:FEAA:BB44
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Server0
Physical
Config
Services
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 3000:159:1::7
Pinging 3000:159:1::7 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 3000:159:1::7:
Packets: Sent = 4, Received = 0, Lost = 4 (100% Loss),
C:\>ping 3000:159:1::7
Pinging 3000:159:1::7 with 32 bytes of data:
Reply from 3000:159:1::7: bytes=32 time=0ms TTL=128
Reply from 3000:159:1::7: bytes=32 time=2ms TTL=128
Reply from 3000:159:1::7: bytes=32 time=2ms TTL=128
Reply from 3000:159:1::7: bytes=32 time=7ms TTL=128
Ping statistics for 3000:159:1::7:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 7ms, Average = 2ms
C:\>
Top

8-


Laptop1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::8
/ 64
Link Local Address
FE80::290:21FF:FE7C:8C41
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top



- Conclusion :
  Ce TD m'a permis de comprendre les différents types de configurations même si c en IPV4 ou IPV6.

  Sur les photos vous pouvez voir qu'il y a 2 configurations dans les commandes c'est parce que j'ai pas prêtée assez d'attention aux configurations.